

Executive Summary – Fashion-MNIST Classification Using Neural Networks

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This study focuses on developing and evaluating two types of neural network models — an Artificial Neural Network (ANN) and a Convolutional Neural Network (CNN) — to perform image classification on the Fashion-MNIST dataset. The dataset comprises 60,000 training and 10,000 testing grayscale images representing ten categories of clothing items, such as T-shirts/tops, trousers, pullovers, dresses, coats, sandals, shirts, sneakers, bags, and ankle boots.

Model Architectures:

The **ANN** model is built with two fully connected hidden layers containing 128 and 64 neurons, each activated using the ReLU function. The output layer consists of 10 neurons with a softmax activation to classify the images into ten distinct categories.

- The **CNN** model comprises two convolutional layers with 32 and 64 filters, respectively, followed by max-pooling layers for spatial reduction. The extracted features are then passed through a dense layer of 100 neurons before reaching the final softmax output layer.

Both models were optimized using the **Adam optimizer** and trained with the **sparse categorical cross-entropy loss function** for ten epochs, utilizing 10% of the training data for validation.

Model	Test Accuracy	Test Loss
ANN (MLP)	88.29%	0.3516
CNN	91.3%	0.2728

Findings and Analysis:

The results show that the **CNN** achieved a higher test accuracy than the **ANN**, outperforming it by approximately **2.8%**. This improvement highlights the CNN's superior ability to capture spatial and structural features within image data through convolutional layers. Both models performed well overall; however, misclassifications were observed among visually similar clothing types, particularly between **Shirts** and **Coats**, due to their comparable textures and shapes. The confusion matrix further indicated strong recognition accuracy for clearly distinct items such as **Sandals** and **Sneakers**, confirming the CNN's robustness in feature extraction and classification.

The training and validation accuracy and loss curves confirmed consistent convergence and stable learning for both models, indicating that neither suffered from significant overfitting or underfitting.

Potential Improvements:

Future work could focus on enhancing model performance and generalization. Techniques such as **data augmentation** can be used to artificially expand the dataset, introducing more variations and reducing overfitting. Incorporating **dropout layers** may further improve robustness by preventing the network from relying too heavily on specific neurons. Additionally, fine-tuning the **optimizer parameters** or implementing a **learning rate scheduler** could lead to faster and more stable convergence. Exploring deeper CNN architectures or applying **transfer learning** with pre-trained models like **VGG16** may also help achieve accuracy levels exceeding 93%.

Conclusion:

This project successfully achieved its objectives by designing, training, and evaluating two neural network architectures for the Fashion-MNIST image classification task. The **Convolutional Neural Network (CNN)** demonstrated higher accuracy and stronger generalization compared to the **Artificial Neural Network (ANN)**, confirming its effectiveness for visual pattern recognition. Through this work, key concepts such as data preprocessing, model construction, optimization, and evaluation were effectively applied. Overall, the study reinforces the value of CNNs in handling image-based problems and provides a strong foundation for future exploration of more advanced deep learning models.